

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO -2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Design Task

Weightage: 20%

QUESTIONS:

Total Marks: 20

Question 1: Responsive Navigation Bar Design - Marks(4)

Suppose that an e-commerce website requires a responsive navigation bar that adapts across desktop, tablet, and mobile devices.

- (a) Identify key components required in a navigation bar for an e-commerce website.
 - (b) Design a responsive navigation bar using appropriate CSS techniques.
 - (c) Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).
 - (d) Evaluate the usability and suggest improvements for better user experience.
-

Question 2: Product Hover Effect - Marks(4)

Suppose that an e-commerce website wants to enhance user interaction through visual effects on product items.

- (a) Identify suitable CSS properties for hover effects.
 - (b) Suggest appropriate **CSS animation or transition** for product interaction.
 - (c) Design a hover effect for a product card.
 - (d) Evaluate performance and user experience considerations.
-

Question 3: Layout Selection (Flexbox vs Grid) - Marks(4)

Suppose that a dashboard layout needs to be designed for a web application.

- (a) Identify layout requirements of a dashboard (rows, columns, alignment).

- (b) Compare **Flexbox and Grid** based on layout needs.
 - (c) Choose the most suitable layout technique and justify your choice.
 - (d) Evaluate the scalability and responsiveness of the chosen approach.
-

Question 4: Mobile-First Blog Layout - Marks(4)

Suppose that a blog page must be designed following a mobile-first approach.

- (a) Identify key sections of a blog layout.
 - (b) Design a mobile-first layout using CSS.
 - (c) Demonstrate how the layout scales for larger screens using media queries.
 - (d) Evaluate accessibility and readability of the design.
-

Question 5: JavaScript Event Handling for Dynamic Menu - Marks(4)

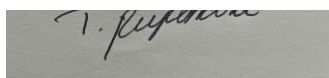
Suppose that a website includes a dynamic menu that responds to user interactions.

- (a) Identify appropriate JavaScript events for menu interaction.
- (b) Design event handling logic for menu toggle functionality.
- (c) Implement interaction for user actions (click/hover).
- (d) Evaluate performance and suggest improvements.

Code of Conduct:

I T Rupendra/(192311325) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Concept	25	Demonstrates complete understanding of design requirements and concepts	Good understanding with minor gaps	Basic understanding with some errors	Poor understanding or incorrect concepts
Design / Implementation	30	Well-structured, efficient, and responsive	Correct implementation with minor issues	Basic implementation with	Incorrect or incomplete implementation

		design implementation		inconsistencies	
Use of Tools (CSS/JS)	20	Effective and advanced use of tools/techniques	Appropriate use of tools	Limited or inconsistent use	Incorrect or minimal use
Analysis & Evaluation	15	Insightful analysis with strong justification and optimization	Good analysis with justification	Basic analysis with limited depth	Poor or no analysis
Documentation / Clarity	10	Clear, well-structured, and properly presented solution	Mostly clear with minor issues	Basic clarity, missing details	Poor or unclear documentation

Question 1: Responsive Navigation Bar Design (4 Marks)

(a) HTML

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Responsive Navbar</title>
<link rel="stylesheet" href="style.css">
</head>
<body>

<nav class="navbar">
  <div class="logo">ShopEasy</div>

  <ul class="nav-links" id="menu">
    <li><a href="#">Home</a></li>
    <li><a href="#">Products</a></li>
    <li><a href="#">Categories</a></li>
    <li><a href="#">Cart</a></li>
    <li><a href="#">Contact</a></li>
  </ul>

  <div class="menu-icon" onclick="toggleMenu()">≡</div>
</nav>

<script>
function toggleMenu(){
  document.getElementById("menu").classList.toggle("show");
}
</script>

</body>
</html>
(b,c) CSS
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
}

.navbar{
  display:flex;
  justify-content:space-between;
```

```
    align-items:center;
    background:#333;
    color:white;
    padding:15px;
}

.logo{
    font-size:24px;
    font-weight:bold;
}

.nav-links{
    display:flex;
    list-style:none;
}

.nav-links li{
    margin:0 15px;
}

.nav-links a{
    color:white;
    text-decoration:none;
}

.menu-icon{
    display:none;
    font-size:30px;
    cursor:pointer;
}

@media(max-width:768px){
    .nav-links{
        display:none;
        flex-direction:column;
        width:100%;
        background:#444;
    }

    .nav-links.show{
        display:flex;
    }

    .menu-icon{
```

```
        display:block;
    }
}
```

(d)

- Easy navigation on all devices.
- Hamburger menu saves space.
- Add search bar and cart icon for better UX.

Question 2: Product Hover Effect (4 Marks)

HTML

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" href="style.css">
</head>
<body>

<div class="card">
    
    <h3>Smart Watch</h3>
    <p>₹2999</p>
</div>

</body>
</html>
```

CSS

```
.card{
    width:220px;
    text-align:center;
    border:1px solid #ccc;
    padding:15px;
    transition:0.4s;
}

.card:hover{
    transform:scale(1.05);
    box-shadow:0 0 15px gray;
    cursor:pointer;
}
```

Evaluation

- Smooth transition improves interaction.
 - Uses transform for better performance.
 - Avoid excessive animations.
-

Question 3: Layout Selection (Flexbox vs Grid) (4 Marks)

HTML

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" href="style.css">
</head>
<body>

<div class="dashboard">
  <div>Header</div>
  <div>Sidebar</div>
  <div>Main Content</div>
  <div>Footer</div>
</div>

</body>
</html>
```

CSS (Grid Layout)

```
.dashboard{
  display:grid;
  grid-template-columns:200px 1fr;
  grid-template-rows:70px 300px 60px;
  gap:10px;
}

.dashboard div{
  background:lightblue;
  padding:20px;
}

.dashboard div:nth-child(1){
  grid-column:1/3;
}

.dashboard div:nth-child(4){
  grid-column:1/3;
}

@media(max-width:768px){
  .dashboard{
    grid-template-columns:1fr;
  }
}
```

```
.dashboard div{
  grid-column:auto;
}
}
```

Evaluation

- Grid is best for dashboard layouts.
- Easy to create rows and columns.
- Responsive using media queries.

Question 4: Mobile-First Blog Layout (4 Marks)

HTML

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" href="style.css">
</head>
<body>

<header>My Blog</header>

<main>
  <article>
    <h2>Blog Title</h2>
    <p>Blog content goes here...</p>
  </article>

  <aside>
    Sidebar
  </aside>
</main>

</body>
</html>
```

CSS

```
body{
  font-family:Arial;
  margin:0;
}

header{
  background:#333;
  color:white;
  padding:15px;
}
```



```

main{
  display:block;
  padding:15px;
}

article,aside{
  background:#eee;
  padding:15px;
  margin-bottom:10px;
}

@media(min-width:768px){
  main{
    display:flex;
    gap:20px;
  }

  article{
    flex:3;
  }

  aside{
    flex:1;
  }
}

```

Evaluation

- Mobile-first approach improves responsiveness.
- Readable content with proper spacing.
- Accessible on all screen sizes.

Question 5: JavaScript Event Handling for Dynamic Menu (4 Marks)

HTML

```

<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" href="style.css">
</head>
<body>

<button id="btn">Menu</button>

<ul id="menu">
  <li>Home</li>

```

```
<li>About</li>
<li>Services</li>
<li>Contact</li>
</ul>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

CSS

```
#menu{
    display:none;
    list-style:none;
    background:#333;
    color:white;
    padding:10px;
}
```

```
.show{
    display:block;
}
```

JavaScript

```
let btn=document.getElementById("btn");
let menu=document.getElementById("menu");

btn.addEventListener("click",function(){
    menu.classList.toggle("show");
});

menu.addEventListener("mouseover",function(){
    menu.style.background="darkblue";
});

menu.addEventListener("mouseout",function(){
    menu.style.background="#333";
});
```